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SETTLEMENT OF SPINY LOBSTER, *PANULIRUS ARGUS* (LATREILLE, 1804), IN FLORIDA: PATTERN WITHOUT PREDICTABILITY?

BY

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ABSTRACT

We used plankton nets, floating postlarval collectors, and arrays of benthic settling devices, along with diver surveys of juvenile lobster abundance and nursery habitat structure, to estimate the spatial pattern of settlement, abundance of settlers, and characteristics of postsettlement juvenile *Panulirus argus* populations in Florida Bay, the primary nursery for spiny lobsters in Florida. Within a 200 km^2 region of Florida Bay, settlement was patchy and locally unpredictable, although settlement occurred at most sites each lunar phase. However, the number of postlarvae entering inlets to the bay was significantly correlated with regional settlement, and areas with abundant red macroalgae (settlement substrate) and numerous sponges (benthic juvenile shelter) were the most productive sites, even though settlement within them varied widely during lunar influxes. Floating collector catches accurately estimated the number of postlarvae in the water column at inlets, but results from collectors deployed in the bay did not correlate with the number of postlarvae settling on benthic collectors nearby. Estimates of postsettlement mortality in the field yield a natural mortality of about 97% in the year following settlement.

RÉSUMÉ

Afin d'estimer la répartition spatiale de la sédentarisation, l'abondance des animaux sédentarisés, et les caractéristiques des populations de juvéniles *Panulirus argus* sédentarisés dans la baie de Floride, la principale nursery de langoustes en Floride, nous avons utilisé des filets à plancton, des collecteurs flottants de post-larves, et des batteries d'engins de prélèvements benthiques, effectuant parallèlement des contrôles, en plongée, de l'abondance des juvéniles de langouste et de la structure de l'habitat nursery. A l'intérieur d'une zone de 200 km^2 dans la baie de Floride, la sédentarisation était inégale et imprévisible localement, bien qu'elle survînt dans la plupart des sites à chaque phase lunaire. Cependant, le nombre de postlarves entrant dans les criques de la baie était corrélé significativement avec la sédentarisation dans cette région, et les zones possédant d'abondantes macroalgues rouges (substrat de sédentarisation) et de nombreuses éponges (abri benthique des juvéniles) étaient les sites les plus productifs, même si la sédentarisation y variait largement durant les phases lunaires. Les prises aux collecteurs flottants ont permis d'estimer avec précision le nombre de postlarves dans la colonne d'eau à l'intérieur des criques, mais les résultats obtenus à partir des collecteurs déployés dans la baie ne présentaient pas de corrélation avec le nombre de postlarves recueillies par les collecteurs benthiques voisins. Les estimations sur le terrain de la mortalité après sédentarisation ont abouti à une mortalité naturelle d'environ 97% dans l'année suivant la sédentarisation.

INTRODUCTION

For lobsters, both spiny and clawed, mortality during settlement and the early benthic period is potentially a significant determinant of eventual population distribution and size (Caddy & Stamatopoulos, 1990; Wahle & Steneck, 1991; Jernakoff, 1990; Smith & Herrnkind, 1992). Insufficient postsettlement habitat, with concomitantly high predation rates, is thought to limit juvenile abundance irrespective of settlement number, creating a demographic "bottleneck" (Wahle & Steneck, 1991; Butler & Herrnkind, in press; Parrish & Polovina, in press). If so, population distributions established at settlement will bear little resemblance to later juvenile distributions. Thus, monitoring only the magnitude of arriving settlers (e.g., postlarval catch on artificial collectors) under these circumstances would yield inaccurate estimates of recruitment and fishery potential.

Florida Bay is the primary nursery for south Florida's (U.S.A.) spiny lobster populations. Postlarval Caribbean spiny lobsters, *Panulirus argus* (Latreille, 1804), flush into Florida Bay through interisland channels from the Atlantic Ocean and these influxes peak monthly around new moon (Little, 1977; Little & Milano, 1980), although pulses of postlarvae occasionally arrive at other times (Marx, 1986). The influx of postlarvae often peaks seasonally in the spring and fall, although there can be substantial regional and temporal variability (Marx, 1986). Postlarvae are funneled through the narrow inlets between the Florida Keys as they are carried into Florida Bay by swift tidal currents. The currents eventually slow as they mix and spread about the shallow inner bay (1-3 m depth) dispersing postlarvae, which presumably drift or swim until they settle in macroalgae (especially red algae, such as *Laurencia* spp., Marx & Herrnkind, 1985a; Herrnkind & Butler, 1986). Postlarvae settle within 3-4 days and typically metamorphose within 10 days after entering the bay during summer (Butler & Herrnkind, 1991); metamorphosis may take up to three weeks during winter (Field & Butler, in press). Several months after settling, juveniles leave the algae to reside by day beneath sponges, octocorals, seagrass undercuts and rock crevices (hereafter, these juveniles will be referred to as "post-algal stage juveniles"). Because they are somewhat less cryptic, we find that these post-algal juveniles can be sampled by intensive diver search (Herrnkind et al., 1988).

Sampling using floating collectors has provided extensive information on spiny lobster postlarval influx rates and timing for several species (Herrnkind et al., in press) and such information has been used in western Australia to predict harvestable stocks of *Panulirus cygnus* George, 1962 (cf. Phillips, 1986; Caputi et al., in press). Benthic collecting devices, designed to mimic settlement habitat, have been used to capture settling postlarvae of *Jasus edwardsii* (Hutton, 1875) (cf. Booth, 1979; Hayakawa et al., 1990), *J. novaehollandiae* Holthuis, 1963 (cf. Lewis, 1979) and *P. argus* (cf. Butler & Herrnkind, 1992a). However, quantitative studies on spatial patterns of natural settlement have been constrained by the scarceness, small size, and crypticity of settling and early benthic stage lobsters in structurally complex habitat (Marx, 1986; Jernakoff, 1990; Yoshimura & Yamakawa, 1988).

In the Caribbean, Witham-type floating collectors (Witham et al., 1964) have been used to monitor the influx of *P. argus* postlarvae, but the relationship between the number of postlarvae found on these collectors and actual planktonic abundances is untested. One aspect of our study was to evaluate the validity of postlarval influx estimates based on Witham collector data. The relevance of postlarval influx, as measured on Witham collectors, to subsequent postlarval settlement and recruitment to the benthic juvenile stage is also uncertain. To address this question, we compared settlement of postlarvae in Witham collectors and benthic artificial collectors to the size structure and abundance of recently recruited juveniles in adjacent macroalgal habitat, and we quantified habitat structure at those sites to evaluate relationships among postlarval availability, postlarval settlement, juveniles recruitment, and habitat type. We also deployed arrays of benthic collectors in order to investigate the small- and large-scale spatial patterns of postlarval settlement in Florida Bay.

METHODS

The middle Florida Keys region from Grassy Key to Lower Matecumbe Key includes four inlets into Florida Bay, each of which was monitored for 1-3 lunar periods during the summers of 1988 through 1990 (fig. 1). At each inlet we deployed floating Witham collectors on the oceanside and also at three downstream bayside sites where we set out arrays of benthic collectors as well. In addition, we quantitatively sampled postlarvae using plankton nets equipped with flow meters at inlets during lunar periods of peak influx. At the end of each monitoring period we made a timed diver search of postsettlement habitat near the bayside collector sites to sample locally recruited post-algal stage juveniles, and we also quantified the abundance of various structural features within nursery habitat at each site. These data were used to estimate: (1) postlarval influx (oceanside Witham collectors and plankton nets); (2) the distribution and abundance of subsequent downcurrent settlement (benthic collector arrays); (3) the comparative numbers of surface swimming or drifting postlarvae in the bay (bayside Witham collectors); (4) recent local recruitment to the post-algal juvenile stage (juvenile abundance and size structure); and (5) the relationship among levels of postlarval settlement, juvenile recruitment, and various habitat features.

Site selection and sampling schedule

We initially monitored the Tom's Harbor channel during June and July 1988. Thereafter, we monitored that site and one other channel location each lunar period (June, July, August 1989, June, July 1990; see fig. 1). In all, we monitored the Tom's Harbor channel during seven lunar periods and five other channels once each. Surface current tracks were measured on an incoming tide at full moon prior to a sampling period to approximate the tidal currents at new moon. Twelve individually numbered current markers (12 cm dia. grapefruit), the tops of which barely break the water surface, were deployed in sets of three approximately every 100 m across a channel. The current markers were set out just before peak flood tide then tracked by boat using Loran to determine positions (nearest hundredth of a minute of latitude and longitude) every 20 minutes until drifting stopped. Water depth seldom exceeded 3 m and diver observations of subsurface drifting debris confirmed that water flow directions

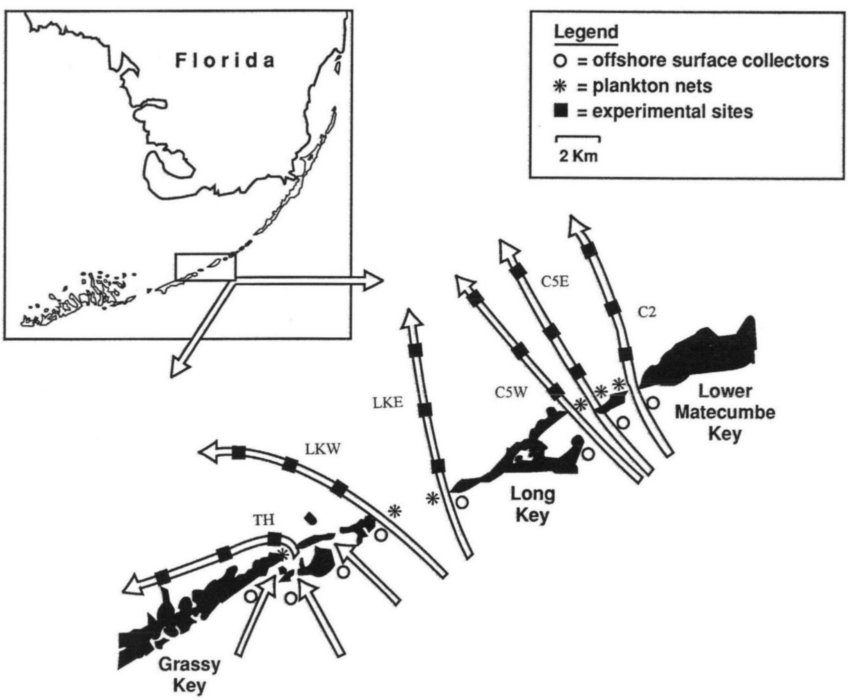

Fig. 1. Diagram of sampling locations and measured paths of flooding tidal currents in central Florida Bay. Postlarvae entering the bay were sampled by Witham surface collectors (O) on the oceanside of each inlet as well as by plankton nets (*) at bridges over inlets. At three stations along each current track (squares), postlarvae were sampled by Witham surface collectors and settlement on the bottom was sampled by arrays of benthic collectors. Near each station, natural nursery habitat was characterized and recently-recruited juveniles were censused. Inlets: TH, Tom's Harbor; LKW, Long Key West; LKE, Long Key East; C5W, Channel 5 West; C5E, Channel 5 East; C2, Channel 2.

were equivalent throughout the water column. We then established three downstream, bayside monitoring sites near natural juvenile habitat (hardbottom areas dominated by algae, sponges and octocorals) at approximately 1-3 km intervals along the center of each tidal track. This assured that water from one inlet flowed over the sites despite any deflection caused by variable winds.

Estimating postlarval influx

We estimated the relative numbers of postlarvae entering each inlet nightly by monitoring floating (Witham-type) collectors (Herrnkind et al., 1988; Heatwole et al., 1991; Butler & Herrnkind, 1992b) and by fishing plankton nets from bridges over the inlets on nighttime flooding tides. Six Witham collectors were set out (each approx. 10-15 m apart) across current 0.1-1.0 km seaward of each inlet, depending on topography of a site. We sampled the collectors daily for seven days, beginning three days prior to new moon and recorded the number of postlarvae on each. To directly estimate postlarval abundance, and for comparison with Witham collector catch, we sampled postlarvae at each inlet using 1 m dia., 750 μm mesh plankton nets equipped with flow meters. Four 20 min. samples were taken over a 2 hr. period each night during peak tidal inflow for five nights beginning one day before new moon. The net tops were set at 0.5-1.0 m below the surface where postlarvae are concentrated (Little, 1977; Little & Milano, 1980); inlet depths ranged from 1.8 to 3.6 m.

Estimating postlarval settlement

Natural benthic collectors (NBC's) consisted of 37 liter plastic mesh (1 cm square) bags containing, freshly collected algae, (*Laurencia* spp.); each collector held the equivalent of a dense algal clump approx. 50 cm dia. Each mesh bag was slit many times so algal fronds protruded; the narrow slits and open mesh permitted entry by postlarvae but obstructed access by most predatory fish and crabs. Rectangular arrays of NBC's consisted of 25 (5 × 5 array, 150 m^2 area), or 12 (4 × 3 array, 75 m^2 area) collectors set 2-3 m apart and anchored by a rock or a PVC stake driven into the substrate. Styrofoam floats attached to the top of each NBC provided buoyancy and kept the collector upright like natural algae. We also employed artificial benthic collectors (ABC's) of folded filter material preconditioned by soaking at dockside for 2 weeks. Each ABC consisted of a 4.5 m long × 20 cm wide × 4 cm thick section of filter material (the same as that used in Witham-style collectors) folded accordion-style within a slitted plastic mesh bag with Styrofoam floats and a PVC anchor stake. To compare catch rates of NBC's and ABC's, we used separate 5 × 5 NBC and ABC arrays in 1989 and alternated each collector type in 4 × 3 arrays in 1990.

The benthic collector arrays (of both types) were placed on patches of sparse turtle grass (*Thalassia testudinum* Koenig), a poor settlement substrate, near areas of hardbottom nursery habitat in depths ranging from 1.5-2.5 m (Herrnkind &

Butler, 1986; Sweat, 1968). The three downstream stations were 0.6-2.7 km, 2.0-5.2 km, and 6.2-8.1 km into the bay, respectively; the exact location of each station downstream of an inlet depended on the current path and location of suitable substrate. In 1988 and 1989, one 5 × 5 array was placed at each of the three monitoring sites along the pathway. In 1990, we used two 4 × 3 arrays placed across current (approximately 200 m apart) to estimate settlement variability at a spatial range intermediate between the meter-scale intercollector distance and kilometer-scale interstation distances. Because the arrays were uniform, we assume they measure the number of imminently settling postlarvae that encounter the benthic collectors independent of location in the bay.

The benthic arrays were set out three days prior to new moon and collected 7-10 days later to include the peak settlement period. The collectors were retrieved by divers then immediately searched. Double-blind tests of searching accuracy in NBC's were performed prior to each collection (mean = 67% accuracy, n = 39). We recorded the number, developmental stage, size, and collector position in the array for each settler. Along with benthic collectors, we placed 4-5 floating Witham collectors near each station to monitor planktonic postlarval abundance. These were monitored daily for seven consecutive days beginning three days before new moon to include the peak influx period.

Juvenile lobster and habitat sampling

We estimated recent local recruitment to the juvenile stage by diver sampling of post-algal stage juveniles (approx. 15-45 mm carapace length; CL) within approximately 100 m radius of each benthic collector array. Because post-algal stage juveniles are diurnal crevice-dwellers, they can be visually censused whereas algal-dwelling juveniles cannot. We calculated catch per unit effort (CPUE) based on the number of juveniles found under typical juvenile shelter (e.g., sponges, octocorals, and rock holes) in 30 minute searches by two divers (i.e., 1 diver hour/site; Herrnkind, et al., 1988). Postalgal lobsters are less nomadic than large juveniles or adults and were likely to have settled in the immediate area (Andree, 1981; Butler & Herrnkind, in press; Forcucci et al., in press). Therefore, the abundance and size distribution of post-algal stage juveniles at each site represented an estimate of local recruitment inclusive of both settlement and postsettlement processes (i.e., mortality and growth). We used line transects (6-25 m long; n = 3-6 per site) to estimate percent substrate cover including: algae, sand, seagrass, sponges, and octocorals. All features over 20 cm long were recorded.

RESULTS

Collector comparisons

Catch rates of both types of collectors were compared as separate 5 × 5 arrays (1989) and when interspersed in 4 × 3 arrays (1990). In both cases, mean catch rates for ABC's were slightly higher than for NBC's although high variance made them statistically indistinguishable. In 1989, we collected a mean of 0.05 postlarvae per NBC/d (i.e., per day) versus 0.08 postlarvae per ABC/d (1-Factor ANOVA on ln transformed data; $F = 1.48$; $df = 1, 14$; $p>0.05$), whereas in 1990 we collected, on average, 0.04 postlarvae per NBC/d versus 0.06 postlarvae per ABC/d ($F = 4.83$; $df = 1, 46$; $p>0.05$). Laboratory experiments testing postlarval selection of settlement substrates (see methods in Herrnkind & Butler, 1986) indicated that there was no significant difference in postlarval choice of artificial material versus freshly collected macroalgae of equal volume (log-linear frequency table analysis; $G = 1.03$; $df = 1$; $p>0.05$; $n = 36$ trials). We also compared the catch rate of NBCs with natural, undisturbed algal clumps by counting newly settled postlarvae in equivalent amounts of freshly collected macroalgae near a NBC array (Tom's Harbor station 3) in July 1989. Twenty-five NBCs yielded 19 settlers (0.72 postlarvae/collector) while fresh algae equal to half as many (twelve) NBCs yielded 8 settlers (0.66 postlarvae/clump); that is, equivalent numbers of settlers for a given volume of algae.

Patterns of postlarval influx

Samples from oceanside Witham collectors and plankton net tows at each inlet were compared by Pearson correlation and t-tests of significance (ln transformed data) both for the same day and over a total lunar sampling period (table I). Daily Witham collector catches ranged from 0-16 postlarvae per collector (mean = 0.57/collector, n = 48 collections) while net catches ranged from 0-54 postlarvae per 20 min. tow (mean = 4.4/tow, n = 194 casts). Nightly postlarval densities in each inlet (calculated from plankton net sample volume) ranged from 0-0.16 postlarvae per m^3 (mean = $0.01/m^3$, n = 48 sampling nights).

Data obtained from both sampling techniques over full lunar periods showed high variation among inlets (i.e., lunar postlarval influx differed between inlets except during June 1989; table I). Similarly, there was great temporal variation in postlarval influx at the Tom's Harbor inlet, which was repeatedly sampled over the three summers (range 0.002-0.03 postlarvae/m^3, n = 7). The plankton and Witham collector catch rates for each lunar period at a channel were strongly correlated ($r = 0.75$, $p = 0.005$, $n = 12$; fig. 2, top panel). Daily correlations were statistically significant but substantially weaker ($r = 0.38$, $p = 0.008$, $n = 48$). Thus, both sampling methods provided comparable estimates of the relative magnitude of postlarval influx. The lower correlation for the same-day comparisons suggests that small-scale temporal and spatial variation in postlarval densities may be substantial.

Settlement patterns

Plankton net and oceanside Witham collector samples for each channel and lunar period were also independently examined for correlations with the level of settlement in benthic collectors at the downstream sites using Pearson correlation and t-tests of significance. Plankton net catch was weakly correlated with postlarval settlement at downstream sites (fig. 2, middle panel; $r = 0.53$, $p = 0.08$; n = 12). That is, the magnitude of an influx at a given inlet over a lunar period was roughly reflected in the total number of postlarvae settling in the region downstream of the inlet. In contrast, the number of postlarvae caught on oceanside Witham collectors was uncorrelated with postlarval settlement in the bay downstream of the inlet ($r = 0.09$; $p > 0.05$; n = 47). Bayside Witham collector catches were uncorrelated with benthic collector catches at the same station during all sampling periods ($r = 0.02$, $p = 0.89$, n = 47). These results collectively suggest that the relative numbers of postlarvae near the surface in the bay, and which alighted and remained on surface collectors, did not accurately reflect either the numbers entering the inlet the previous night or the numbers settling in benthic substrates.

Postlarval settlement in the benthic collector arrays at downstream sites was likewise uncorrelated with distance from an inlet (fig. 2, bottom panel; $r = -0.18$; $p > 0.05$; n = 47); settlement patterns were inconsistent among inlets and from month to month at a single site. For example, the Tom's Harbor inlet was sampled over three summers and seven lunar periods, but exhibited no repetitive pattern despite similar current flow rates, current pathways, and settlement habitat distribution. Over the distances we sampled, postlarval settlement was distinctly patchy (fig. 3).

We applied a 5-factor unbalanced hierarchical ANOVA to the benthic collector data to determine the spatial and temporal scales at which settlement was most variable (table II). In the analysis, we considered spatial variation in settlement among inlets, sites along a current path and separated by 2-4 km, stations separated by 100-200 m at a site, and finally among individual benthic collectors separated by 1 m within arrays at each station. We also considered temporal variation in postlarval settlement among years and months at a single inlet, Tom's Harbor. Intercollector variation in settler number accounted for most of the variance (80.4%, $p < 0.01$), followed by differences among inlets (10%, $p < 0.01$), and sites along a single current path (7.1%, $p < 0.01$; table II). Stations situated 100-200 m apart had similar settlement levels (0%, $p > 0.05$) and differences among years (0%, $p > 0.05$) and months (2.5%, $p > 0.05$) accounted for little of the overall variance in the pattern of postlarval settlement.

Settlement of postlarvae within the benthic collector arrays was not significantly different from random ($p > 0.05$; standardized Morisita's Index of Dispersion: -0.22 to 0.46) in 23 of 24 NBC arrays in which two or more postlarvae settled (range 2-29 postlarvae/array). Postlarvae typically settled alone within NBCs, although multiple postlarvae (up to 5 per NBC) settled on 2.2% of all NBCs. Only during the highest recorded settlement event (July 1988, Toms Harbor site 1, n= 30 collectors, mean = 0.97 settlers/collector) was settlement of postlarvae within an NBC array significantly clumped ($p < 0.05$; Standardized

TABLE I

Data on postlarval abundance, settlement, and juvenile spiny lobster, *Panulirus argus* (Latreille), population and habitat characteristics at sites in Florida Bay during the summers of 1988-1990. Table heading definitions: SCO, mean daily number of postlarvae on oceanside surface (Witham) collectors; Net, mean number of postlarvae per m^3 from nightly plankton net tows; Site/Station, sites located downcurrent from the inlet named and stations within a site (labeled A and B); Dist., distance from inlet to site; SCB, mean daily number of postlarvae on bayside surface (Witham) collectors at a site or station; BC, mean number of newly settled postlarvae and first-stage juveniles per benthic collector at a site or station (data from artificial and natural benthic collectors pooled); Algae, mean percent bottom coverage by red algae (*Laurencia* spp.) at a site or station; CPUE, the number of postalgal stage juveniles lobsters (<45 mm CL) captured per hour of dive time at a site or station; CL, the mean carapace length of lobsters captured at a site or station; d, dive; coll., collector; na, not available.

Inlet	Date	SCO (no./d/coll.)	Net (no./m³)	Site/ Station	Dist. (km)	SCB (no./d/coll.)	BC (no./d/coll.)	Algae (%)	CPUE (no./hr)	CL (mm)
Tom's	vi/88	0.31	0.025	0	1.1	0.20	0.08	na	7	33.0
Harbor				1	1.5	0.07	0.39	na	na	na
	vii/88	0.44	0.011	1	1.5	0.11	0.97	na	40	17.9
				2	2.0	0	0.27	na	15	22.5
				3	6.2	0.05	0.10	na	23	23.0
	vi/89	0.28	0.024	1	1.1	0.03	0.13	71	14	17.7
				2	2.9	0	0.13	46	6	17.5
				3	6.2	0.03	0.04	19	15	28.1
	vii/89	0.10	0.002	1	1.1	0.03	0.43	71	2	27.3
				2	2.9	0.11	0.04	46	5	18.6
				3	6.2	0	0	19	9	26.7
	viii/89	0.44	0.030	1	1.1	0.06	0.80	71	4	20.0
				2	2.9	0.08	0.42	46	5	18.6
				3	6.2	0.04	0.40	19	7	26.6

	vi/90	0.27	0.010	1A	1.1	0	0.21	76	10	20.0
				1B	1.2	0	0.10	15	0	na
				2A	2.9	0.14	0.08	12	2	22.2
				2B	2.9	0.14	0.17	21	4	20.0
				3A	6.1	0	0.08	11	8	24.1
				3B	6.2	0	0.25	22	1	24.3
	vii/90	0.18	0.005	1A	1.1	0	0.50	76	9	16.1
				1B	1.2	0	0.08	15	6	21.4
				2A	2.9	0.2	0.25	12	1	12.9
				2B	2.9	0.2	0.42	21	4	22.3
				3A	6.1	0	0.05	11	5	26.4
				3B	6.2	0	0.83	22	4	18.3
Long Key East	vi/89	0.08	0.001	1	1.9	0.09	0.04	0	0	0
				2	4.8	0.03	0.16	0	1	23.2
				3	7.1	0	0.12	9	3	32.8
Channel 2	vii/89	0	0.002	1	2.1	0	0.28	4	6	32.0
				2	5.2	0	0.26	13	3	32.9
				3	8.1	0	0.13	26	5	32.3
Channel 5 West	viii/89	0.68	0.015	1	2.7	0.03	0.31	3	0	0
				2	4.6	0.16	0.53	23	2	30.4
				3	7.2	0.10	0.56	na	na	na
Channel 5 East	vi/90	0.05	0.006	1A	2.0	0	0.25	4	0	0
				1B	2.0	0	0.50	1	1	37.3
				2A	4.4	0.03	0.31	7	0	0
				2B	4.4	0.03	0.74	9	2	24.4
				3A	8.0	0.03	0.10	4	0	0
				3B	8.0	0.03	0.10	4	0	0
Long Key West	vii/90	0.67	0.009	1A	1.8	0	0	22	1	39.0
				1B	1.8	0	0	26	0	0
				2A	3.4	0	0	32	6	27.8
				2B	3.4	0	0.17	31	4	21.8
				3A	8.0	0.03	0.72	31	10	24.9
				3B	8.0	0.03	0.42	36	5	29.3

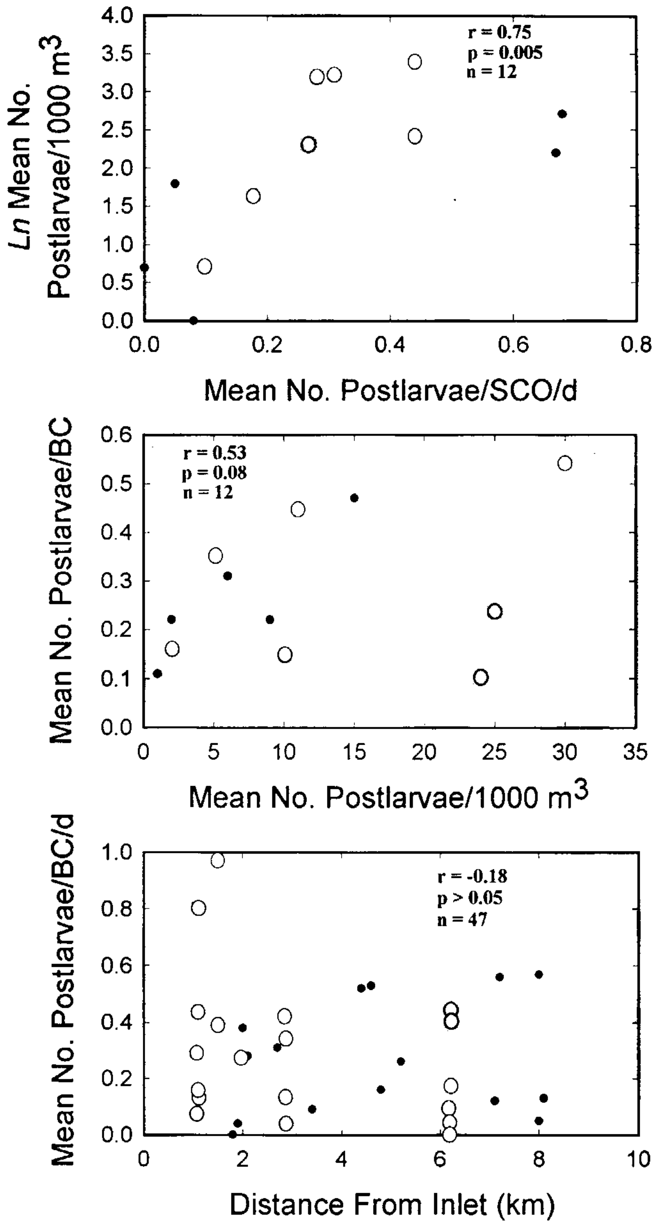

Fig. 2. Top panel: plot of the relationship between plankton net catch of postlarvae at each inlet and nearby oceanside Witham surface collector catch. Data are based on daily catches accumulated over new moon lunar periods between 1988 and 1990. Middle panel: the relationship between the number of postlarval lobsters collected in plankton nets and settlement in benthic collectors in Florida Bay. Bottom panel: plot of the number of postlarvae settling in benthic collectors deployed at different distances from the source inlet. Data collected at the Tom's Harbor inlet and associated bay sites are indicated with open circles and the remaining data with closed circles.

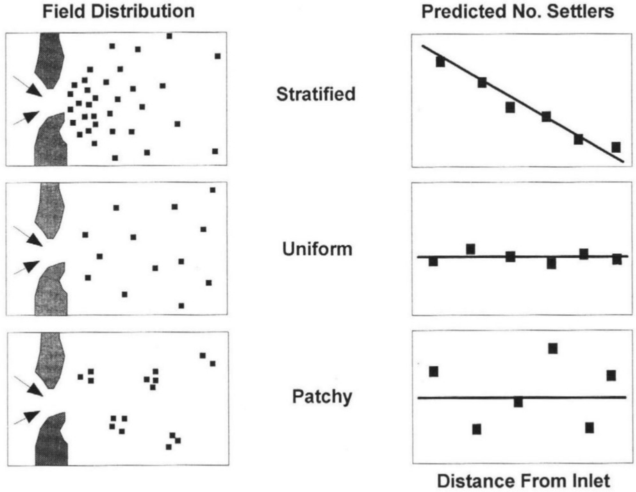

Fig. 3. Hypothetical spatial patterns of spiny lobster postlarval settlement in Florida Bay. Postlarvae are initially concentrated as they funnel through inlets (arrows) between islands (stippled areas). In each panel, the predicted number of settlers (y-axis) is plotted against the distance from the inlet of entry (x-axis). Stratified settlement (top panel): caused by gradual dispersal of postlarvae by spreading currents and rapid settlement. Uniform settlement (middle panel): caused by rapid mixing of currents, undirected swimming of postlarvae, or homogeneous settlement habitat. Patchy settlement (bottom panel): caused by a patchy planktonic distribution of postlarvae that are highly selective of habitat or by heterogeneous settlement habitat.

TABLE II

Results of a five factor unbalanced nested ANOVA examining spatio-temporal patterns of postlarval spiny lobster, *Panulirus argus* (Latreille), settlement in Florida Bay, Florida. In this analysis, variation in postlarval settlement in benthic collectors is partitioned into sources attributable to differences among: (1) years, (2) summer months, (3) inlets leading into Florida Bay, (4) sites downcurrent from the inlets and separated by 2-4 km, (4) stations within sites and separated by 100-200 m, and (6) individual benthic collectors within arrays at each station or site and separated by 1 m.

Source	df	SS	F	P	Percent variance explained
Year	1	0.0002	0	0.9540	0
Month	3	2.7528	13.64	0.0001	2.5
Inlet	5	3.2055	9.54	0.0001	10.0
Site	20	3.0639	2.28	0.0013	7.1
Station	12	0.6860	0.85	0.5989	0
Collectors (error)	541	36.3853			80.4
Total	582	46.0948			

Morisita's Index = 0.51). We used the number of settlers found on benthic collectors and the estimated area of each array to calculate a mean regional density of settlement at each influx of one settler/31m^2 of settlement habitat (maximum of one settler/6.5m^2).

Settlement-recruitment patterns

Finally, we ran a series (yearly data and all years combined) of best-fit multiple regressions in an effort to construct equations that adequately predicted either the magnitude of settlement or numbers of postalgal stage juveniles of three size categories (<25 mm CL, <35 mm CL; or <45 mm CL) using a linear combination of variables including: plankton net catch, oceanside Witham collector catch, bayside Witham collector catch, benthic collector catch (only one category used at a time to insure variable independence when predicting juvenile number), number of postalgal juveniles in one of three size categories at a station (only to predict settlement), distance of a station from an inlet, and the proportion of bottom covered by several substrate types at a station (i.e., sand, algae, seagrass, sponge and octocoral).

All of the regressor variables were non-normally distributed and no transformation was found to alleviate the problem. Algal coverage was also significantly correlated with several other variables, as were the number of postlarvae collected from plankton nets and Witham collectors deployed on the oceanside of the Keys. These conditions violate the assumptions of multiple regression, so we: (1) rank transformed (Imam & Conover, 1979) the regressor variables; (2) omitted algal coverage from consideration; and (3) used either plankton net data or oceanside Witham collector data. Using the entire data set (1988-1991), no model accounted for more than 11% of the variance in postlarval settlement in benthic collectors. Model predictions of recruitment to the postalgal juvenile stage were also poor and only explained 23-39% of the variance in juvenile numbers when all the regressor variables were used. Predictions of the number of small postalgal juveniles <25 mm CL were the most accurate of these models ($R = 0.39$; 7 regressor variables), and in each model, sponge coverage was the most influential variable considered. Models based on data collected in 1989 or 1990 alone explained 42-81% of the variance in settlement or juvenile abundance, but the variables that were most influential in these models were inconsistent from year to year. Thus, no combinations of variables yielded consistently accurate predictions of local settlement or postalgal juvenile abundance (i.e., recruitment).

DISCUSSION

We view these data from two perspectives: what can be interpreted about processes involved in determining the distribution of spiny lobster postlarval settlement within the study area and what might be projected for Florida Bay generally. Hypothetically, postlarvae entering the bay might be expected to settle in a stratified pattern with the highest densities in algal habitat near the inlets where postlarvae are most concentrated, and at succeedingly lower densities farther into the bay (fig. 3). Alternatively, the uneven distribution of planktonic postlarvae, strongly oriented swimming, age-specific settlement, strong substrate selectivity, or hetergeneously distributed settlement habitat would alter this pattern (fig. 3). We found that settlement within a region served by a particular inlet generally reflected the numbers of postlarvae entering the inlet, although settlement within the region was patchy and unpredictable at a given site. Furthermore, some postlarvae settled throughout our study area, which exceeded 200 km^2, at nearly every lunar influx and non-migratory postalgal stage juveniles are found in hardbottom areas farther into Florida Bay than our most distant sites (Holmquist et al., 1989; Field & Butler, in press).

The absence of clear, repeatable settlement patterns within the sampled region reflects high temporal and spatial variability in postlarval abundances and, probably, differences in their readiness to settle (Butler & Herrnkind, 1991), as well as the disruptive mixing of waters from different inlets entering the bay. That postlarvae are heterogeneously distributed is apparent from the low daily correlation between settlement on oceanside Witham collectors and collections from plankton nets deployed nearby. We also commonly noted large differences in successive nightly plankton net catches at an inlet. In fact, the variance in plankton net catches of postlarvae were lowest between nets towed simultaneously within an inlet, whereas variances between successive tows of a single net at the same location in an inlet were as great as those among inlets, months, or years (W. F. Herrnkind and M. J. Butler, unpubl. data). In addition, daily oceanside Witham catches varied greatly among adjacent collectors, indicating uneven, patchy arrival from offshore.

Postlarvae that are ready to settle may swim near the bottom testing substrates, ignoring surface floating collectors, which would explain the poor correspondence between bayside Witham collector catches and postlarval abundance in benthic collectors. At the smallest scale, we found that two or more postlarvae settled on a single benthic collector more frequently than indicated from the field samples of natural algae. Marx (1983) found only one instance of multiple occupancy of a natural algal clump by postlarvae or first benthic stage juveniles. What accounts for this discrepancy? Benthic collectors probably retain most of their settlers because they protect settlers from predators. Under natural conditions, predation may greatly reduce the numbers of the first several benthic juvenile instars, even in algal cover (Smith & Herrnkind, 1992), further contributing to the high dispersion apparent in natural conditions. Moreover, the brief time between settlement and collection probably reduces emigration from benthic collectors because metamorphosis during the summer occurs rapidly, usually within 6-10 days after settlement. Finally, the asocial nature and high vagility of the early benthic stage juveniles (Herrnkind & Butler, 1986) also probably contributes to the small-scale, dispersed distribution of algal-phase juveniles (<20 mm CL) in natural vegetation.

Besides postlarval behavior and transport processes, the location, amount, and character of algal settlement habitat all probably play a role in determining the distribution of regional settlement. Postlarvae avoid heavily silted algae (Herrnkind et al., 1988) and, because they prefer *Laurencia* spp. over seagrass, they settle in highest densities where macroalgal cover is abundant and silt free (Herrnkind & Butler, 1986). However, algal cover and condition vary greatly within Florida Bay. At Tom's Harbor station 1 in June 1990 *Laurencia* spp. covered 76% of the bottom as compared to 4% at Channel 5, station 3B. Algal cover also varies seasonally and from year to year in the same area. In one striking instance, the *Laurencia* at Tom's Harbor Station 1, in July, 1988 was, on average, 30 cm high and covered the seafloor up to the tops of large sponges. A year later (July 1989), it was only 6 cm high, but by the summer of 1990, the algal cover at Tom's Harbor again resembled the conditions there in 1988.

We speculate that large areas of macroalgal substrate also may "filter out" postlarvae carried across them. The effect may explain the different settlement patterns we observed along the Channel 2, Channel 5, Long Key West and Tom's Harbor inlet current tracks. Waters of the former three inlets flow over relatively deep (3-4 m) areas dominated by poor settlement substrate (open sand, sparse seagrasses, and calcareous green algae) within the first 5 km. In these regions, settlement was highest at the most distant bay stations (table I). On the other hand, station 1 at Tom's Harbor (1 km from its inlet), lies next to extensive, shallow meadows of *Laurencia*, wherein we recorded some of our highest settlement readings. Simple explanations of decreasing postlarval settlement with increasing distance into the bay nursery are untenable under such a situation.

The mean density of settlement in our benthic collectors (1 postlarva/31 m^2) is comparable to that estimated previously (1 postlarva/36 m^2; Marx, 1983; Marx & Herrnkind, 1985) from natural macroalgal clumps near our Tom's Harbor station 3. Assuming that the mean estimate of settlement density from all our sites provides a reasonable measure of annual settlement, then we estimate that the region receives 320 settlers/ha each lunar month and 4160 settlers/ha yearly (13 lunar cycles per year). Monthly censuses of postalgal lobsters (15-45 mm CL) by ourselves and others from 1988 to 1992 at a series of sites within three separate regions of Florida Bay, produce a range of mean densities of approximately 50-200 lobsters/ha (Butler & Herrnkind, in press; Forcucci et al., in press; M. J. Butler, W. F. Herrnkind, and J. H. Hunt, unpubl. data). Because it takes about one year for settlers to attain 45-50 mm CL (Forcucci et al., in press; Lellis & Russell, 1990), the postalgal lobsters in an area at any time roughly represent the survivors from the preceding year's settlement. This is true even if some immigrated or emigrated because the densities are derived from repeated samplings at several representative locations. These values yield an estimate of survival from settlement to the postalgal juvenile stage of approx. 3% (an average of 125 postalgal lobsters from 4160 settlers on each hectare of nursery habitat). Recent independent estimates of first year postsettlement survival of spiny lobsters based on mark-recapture of microwire tagged settlers are similar, range 0.6-4.1% (M. J. Butler, J. Hunt, and W. F. Herrnkind, unpubl. data). To our knowledge, these are the first postsettlement mortality estimates for this species based on field sampling of natural populations.

CONCLUSIONS

The relationship between *P. argus* postlarval availability, juvenile abundance, and habitat structure is more complicated than we originally hypothesized, as is demonstrated by the poor correspondence between Witham collector catch and either local settlement, recruitment, habitat, or distance into the bay. We noted similar discrepancies between Witham collector catch and juvenile abundances in a previous study (Herrnkind et al., 1988). In both cases, inappropriate nursery habitat probably precluded settlement or increased postsettlement mortality at our sites, which effectively decoupled postlarval influx from predictions of juvenile recruitment.

The distribution of hardbottom nursery habitat in Florida Bay is not only heterogeneous, but macroalgal (i.e., settlement habitat) distribution within these areas is spatially and temporally ephemeral. When such a shifting mosaic of nursery habitat is overlain by sparse, patchy postlarval settlement, the result is a weak pattern of recruitment without predictability (fig. 4). The specific locations of poor or superior nursery areas change and during each lunar cycle, the arrival of postlarvae at particular locations is highly variable. Under these conditions, settlement and subsequent recruitment of juvenile lobsters in areas of poor habitat will be characteristically low, due to high postsettlement mortality. Postlarvae may or may not arrive in prime nursery areas, so settlement and subsequent recruitment will vary greatly there. Yet, a pattern of recruitment can be distinguished from such a scenario. Poor habitat sites will always have depauperate lobster populations, whereas sites with ample nursery habitat will generally support higher juveniles populations, although the variance in population size in good nursery areas will be large. Due to this variance, recruitment at potentially prime nursery sites, such as those sampled in our study, is unpredictable due to patchy postlarval settlement.

It is a widely held view that the interior of Florida Bay is an essential nursery area for the Florida spiny lobster population, but there is evidence that postlarval transport and survival are limited in this area (Davis & Dodrill, 1980; Holmquist et al., 1989; Field & Butler, in press). If so, the nearshore bay margin nearest the middle Keys probably contributes more to juvenile recruitment in Florida Bay than the inner bay basins to the north. This region is especially

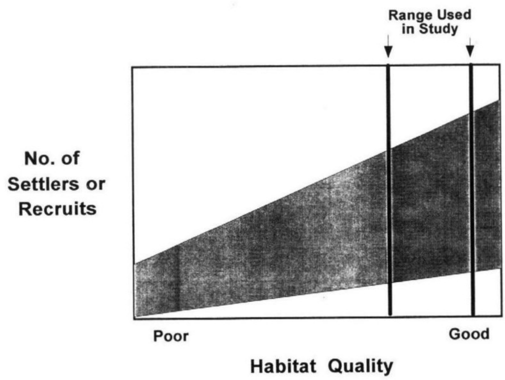

Fig. 4. Conceptual diagram showing the probable relationship between the number of settling postlarvae or juvenile recruits over a range of nursery habitat conditions. The numbers of postlarvae arriving (y axis) within nursery areas of a particular quality (x axis) vary (shaded area), but relatively few postlarvae settle or survive in areas of "poor" habitat (e.g., sand, sparse seagrass or macroalgae). In areas of "good" habitat (e.g., complex hardbottom with extensive macroalgal meadows), postlarval settlement remains variable but survival is high so the mean number of successful recruits and variance in recruitment is greater than in areas of poor habitat. Monitoring settlement or recruitment only within prime nursery areas (dashed lines) will result in an unpredictable relationship between habitat quality and settlement or recruitment.

susceptible to anthropogenic environmental degradation (e.g., siltation, disturbance, and pollution). Recently, major ecosystem disruptions in the central portion of Florida Bay (i.e., massive seagrass die-offs; Robblee et al. 1989) have led to cyanobacterial blooms and the loss of sponges, hence juvenile lobster shelter, throughout central Florida Bay (Butler et al., unpubl. manuscript). These developments further strengthen our contention that the shallow nursery area near the Florida Keys must be recognized both for its contribution to regional recruitment and its ecological fragility.

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